

Skills Summary

Evaluating Functions from Rules, Tables, and Graphs

Use this reference while completing your worksheet or when studying.

Skill 1 — Putting an input into a rule

What the notation means:

$f(3)$ is the *output* of f when the *input* is 3. The letter in the brackets is the number you substitute — it is never a multiplication. To evaluate, replace every x in the rule with the input, in brackets, then simplify.

Example: $f(x) = 5x - 4$. Find $f(3)$.

Step 1. The input is a number and the rule is written in x , so substitute 3 for x and simplify.

Step 2. $f(3) = 5(3) - 4 = 15 - 4 \Rightarrow f(3) = 11$

Try it: $f(x) = 5x - 4$. Find $f(-2)$.

check: -14

Skill 2 — Reading an output out of a table

Rule: a table pairs each input with its output, one column at a time.

To find $g(3)$: look along the input row for 3, then read the number directly below (or beside) it in the output row. No arithmetic is needed — the value is already there.

Example: Find $g(3)$ from this table.

x	1	2	3	4
$g(x)$	6	10	14	18

Step 1. The value is stored in the table, so this is a lookup: find the input column, read the output.

Step 2. The input 3 sits in the third column; the output row holds 14 there. $\Rightarrow g(3) = 14$

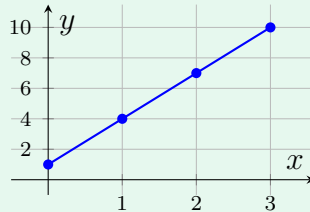
Try it: Find $g(4)$ from the table $x = 0, 2, 4, 6$ with $g(x) = 5, 9, 13, 17$.

check: 18

Skill 3 — Reading an output off a graph

Rule: on a graph, the input is the horizontal coordinate and the output is the vertical one. To find $f(a)$: go to a on the horizontal axis, move up or down to the plotted point, then read across to the vertical axis. That point is $(a, f(a))$.

Example: Find $f(2)$ from this graph.



Step 1. The output is not written anywhere, so read it off the picture: input across, output up.

Step 2. Above $x = 2$ the plotted point is $(2, 7)$, so the output is the second coordinate.
 $\Rightarrow f(2) = 7$

Try it: From the same graph, find $f(3)$.

check: 10

Skill 4 — Comparing two functions at the same input

Rule: you can only compare outputs that come from the *same* input.

Evaluate each function at that input — whichever form it is given in — then compare the two numbers with $<$, $>$, or $=$.

Example: $f(x) = 2x + 5$, and a table gives $h(4) = 12$. Compare $f(4)$ and $h(4)$.

Step 1. The two functions are given in different forms, so evaluate each one separately before comparing.

Step 2. Rule: $f(4) = 2(4) + 5 = 13$. Table: $h(4) = 12$. Now compare 13 with 12.
 $\Rightarrow 13 > 12$

Try it: $f(x) = 2x + 5$ and a table gives $h(2) = 10$. Compare $f(2)$ and $h(2)$.

check: 6 > 10

(!) Watch out: $f(-4)$ and $-f(4)$ are different. In $f(-4)$ the negative goes *into* the rule; in $-f(4)$ you evaluate first and flip the sign afterwards.